TENDER T2 (PF-T200)

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	tender main rail HSS10x4x1/4	2	A500 Gr C	301
2	tender crossmember HSS10x4x1/4	3	A500 Gr C	133
3	tender fish-belly bolster top flange PL12	2	A36	73
4	tender fish-belly bolster web PL10 (deep at centre plate)	4	A36	110
5	tender fish-belly bolster web PL10 (shallow outboard)	8	A36	53
6	tender fish-belly bolster bottom flange PL12 (centre-plate seat)	2	A36	34
7	axle	4	EA4T / 4140	418
8	wheel tread rim 50 deep	8	EN 13262 ER7 / AAR B	651
9	wheel web 25	8	EN 13262 ER7 / AAR B	259
10	wheel hub OD200	8	EN 13262 ER7 / AAR B	165
11	flange	8	EN 13262 ER7	77
12	axlebox	8	purchased	-
13	sideframe PL25	4	A36	240
14	transom HSS10x4x1/4 (upright, pair)	4	A36 / A500 Gr C	195
15	transom top cap PL16	2	A36	29
16	centre plate Ø340 PL12 + Ø60 centre pin	2	A36	17
17	side bearer	4	A36 + UHMW pad	16
18	tender front drop drawbar box PL16	1	A572 Gr 50	142
19	tender front pocket top plate PL20	1	A572 Gr 50	14
20	tender front pocket bottom plate PL20	1	A572 Gr 50	14
21	tender rear drop drawbar box PL16	1	A572 Gr 50	142
22	tender rear pocket top plate PL20	1	A572 Gr 50	14
23	tender rear pocket bottom plate PL20	1	A572 Gr 50	14
24	tender deck 6 mm tread plate (sealed metal spill floor)	1	A36 tread plate	539
25	tender deck cross-bearer C6x8.2 (outboard) web	14	A36	47
26	tender deck cross-bearer C6x8.2 (outboard) top flange	14	A36	23
27	tender deck cross-bearer C6x8.2 (outboard) bottom flange	14	A36	23
28	tender deck cross-bearer C6x8.2 (between rails) web	4	A36	32
29	tender deck cross-bearer C6x8.2 (between rails) top flange	4	A36	16
30	tender deck cross-bearer C6x8.2 (between rails) bottom flange	4	A36	16
31	tender side skin 10 ga (3.4 mm) on posts @500 + 50 mm lining (riveted-plate lo…	2	A1011 CS	-
32	tender end skin 10 ga (3.4 mm) on posts @500 + 50 mm lining	2	A1011 CS	-
33	tender flat roof 10 ga (3.4 mm) on HSS2x2x1/8 bows @500 (sealed)	1	A1011 CS on HSS bows	-
34	tender side post HSS2x2x1/8 @500	20	A500 Gr C	115
35	tender roof bow HSS2x2x1/8 @500	10	A500 Gr C	-
36	tender end post HSS2x2x1/8 @500	8	A500 Gr C	46
37	coal-effect cover top 1.5 mm (16 ga) formed steel, removable (height ASSUMED)	1	A1011 CS 16 ga	-
38	coal-effect cover side skirt 1.5 mm	2	A1011 CS 16 ga	-
39	coal-effect cover end skirt 1.5 mm	2	A1011 CS 16 ga	-
40	tender ENGINE: AIE Kubota V3307-TIE4AB-KEA-1 + radiator + H1P053 pump (rev L4 …	1	purchased	-
41	tender GENERATOR: genset PowerTech PTI-38 open-frame 1708x667x864, 699 kg in r…	1	purchased	-
42	tender GEN-MOUNT: generator mount allowance 80 mm	1	purchased	-
43	tender COMPRESSOR: IR 2340 14 cfm @175 psi, 68 kg, ~3.7 kW, electric clutch, b…	1	purchased	-
44	tender FUEL: custom 150 L 5052 Al diesel tank X3900-4500 in its own compartmen…	1	purchased	-
45	tender FIREWALL: steel firewall plate at X3800, full height, 3 mm A36 (fire se…	1	A36	-
46	tender START-BATT: genset start battery ODX-AGM34 in vented box + master switc…	1	purchased	-
47	tender MAIN-RES: main reservoirs 3 x Midland Haldex 19820 (47 L, 150 psi) long…	1	purchased	-
48	tender RESERVOIR: hydraulic reservoir 600x600x450	1	purchased	-
49	tender COOLER: hydraulic oil cooler >= 14 kW @ 43 C outside air (rev L4), LEFT…	1	purchased	-
50	tender CABINET: RIGHT-wall outward-facing NEMA 3R electrical cabinet: panel QO…	1	purchased	-
51	tender PARTITION: tank-bay partition 50 mm at x2800	1	A36 + insulation	-
52	tender XFMR: shore isolation transformer Acme T2535173S 15 kVA UL 506 NEMA 3R …	1	purchased	-
53	tender XFMR-ISO: 4 elastomer isolators + captive bolts + catcher strap, 3.5 kg…	1	see note	35
54	tender XFMR-CROSS-F: cradle cross angle L3x3x1/4 fwd, rail inner face to inner…	1	A36	59
55	tender XFMR-CROSS-R: cradle cross angle L3x3x1/4 aft, top Z380	1	A36	59
56	tender XFMR-BOXRAIL-1: inrush box rail L3x3x1/4 (288) between the cross angles…	1	A36	13
57	tender XFMR-BOXRAIL-2: inrush box rail L3x3x1/4 (288), right side (TB-3.4)	1	A36	13
58	tender XFMR-HANGER-FL: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
59	tender XFMR-TAB-FL: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
60	tender XFMR-HANGER-FR: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
61	tender XFMR-TAB-FR: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
62	tender XFMR-HANGER-RL: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
63	tender XFMR-TAB-RL: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
64	tender XFMR-HANGER-RR: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
65	tender XFMR-TAB-RR: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
66	tender XFMR-SHIELD: bolted 3 mm steel ballast/spray shield 450 x 1,120 (widene…	1	EN 13262 ER7	-
67	tender INRUSH: inrush limiter (BDESB/MP-ESB-50 class, 120 V coil) in NEMA 4X ~…	1	see note	148

COACH R2 regular (PF-C300) - per coach; accessible coach PF-C300A per CAD

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	coach main rail HSS12x4x1/4	2	A500 Gr C	798
2	end sill HSS12x4x1/4	2	A500 Gr C	128
3	body bolster HSS12x4x1/4 + PL10 doublers	2	A500 Gr C	128
4	axle	4	EA4T / 4140	418
5	wheel tread rim 50 deep	8	EN 13262 ER7 / AAR B	651
6	wheel web 25	8	EN 13262 ER7 / AAR B	259
7	wheel hub OD200	8	EN 13262 ER7 / AAR B	165
8	flange	8	EN 13262 ER7	77
9	axlebox	8	purchased	-
10	sideframe PL25	4	A36	240
11	transom HSS10x4x1/4 (upright, pair)	4	A36 / A500 Gr C	195
12	transom top cap PL16	2	A36	29
13	centre plate Ø340 PL12 + Ø60 centre pin	2	A36	17
14	side bearer	4	A36 + UHMW pad	16
15	crossmember HSS6x2x3/16	10	A500 Gr C	236
16	outrigger HSS6x2x3/16	10	A500 Gr C	70
17	side sill HSS6x2x3/16	2	A500 Gr C	295
18	coach front drop drawbar box PL16	1	A572 Gr 50	161
19	coach front pocket top plate PL20	1	A572 Gr 50	14
20	coach front pocket bottom plate PL20	1	A572 Gr 50	14
21	coach rear drop drawbar box PL16	1	A572 Gr 50	161
22	coach rear pocket top plate PL20	1	A572 Gr 50	14
23	coach rear pocket bottom plate PL20	1	A572 Gr 50	14
24	floor 1 in composite panel	1	purchased	-
25	side wall lower (HSS2x2x1/8 frame + skin)	2	A1011 CS on HSS2x2x1/8	-
26	side wall upper / letter board	2	A1011 CS	-
27	window post HSS2x2x1/8	18	A500 Gr C	-
28	window (polycarbonate)	16	purchased	-
29	end wall (32 in centred door)	4	A1011 CS on HSS	-
30	end wall over door	2	A1011 CS on HSS	-
31	roof eave	1	A1011 CS on HSS bows	-
32	roof crown	1	A1011 CS on HSS bows	-
33	roof clerestory	1	A1011 CS on HSS bows	-
34	platform guard top rail 42 in	8	A53 B sch 40 (2 in, C10d)	16
35	side opening gate bar 42 in (latched, interlocked)	4	A53 B sch 40	18
36	end gate rail	4	A53 B sch 40	82
37	S1 stair stowed - stringer	8	A36 / A500 Gr C (C-304)	-
38	S1 stair stowed - tread	20	A36 / A500 Gr C (C-304)	-
39	S1 stowed handrails (folded flat)	4	A53 B	-
40	S1 vertical swing pivot post 2-1/2 in sch 80 + stowed latch / proximity switch	4	A53 B	76

DRAWBARS / PINS (train set)

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	DB-800 loco-tender drawbar PL32x150 A572-50	1	A572 Gr 50	35
2	DB-800 loco-tender pin Ø60 4140	2	AISI 4140 Q&T	11
3	DB-1473 #1 drawbar PL32x150 A572-50	1	A572 Gr 50	60
4	DB-1473 #1 pin Ø60 4140	2	AISI 4140 Q&T	11
5	DB-1473 #2 drawbar PL32x150 A572-50	1	A572 Gr 50	60
6	DB-1473 #2 pin Ø60 4140	2	AISI 4140 Q&T	11
7	DB-1473 #3 drawbar PL32x150 A572-50	1	A572 Gr 50	60
8	DB-1473 #3 pin Ø60 4140	2	AISI 4140 Q&T	11
9	DB-1473 #4 drawbar PL32x150 A572-50	1	A572 Gr 50	60
10	DB-1473 #4 pin Ø60 4140	2	AISI 4140 Q&T	11

PE STAMP / SIGNATURE

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'-' in kg = purchased, or shell/wall/roof envelope modelled solid in CAD (mass from ledger). Coach R2 adds (not separately modelled): roof bows HSS3x2x1/8 @880 (R1-f), guard posts 2 in sch 40 (C10d), ERV HE05IN roof pod + heat pump per coach (E-701).
Drawgear purchase set (5 joints): 12 pins Ø60 × 150 under head 4140 Q&T (10 + 2 spares) with bushes, Ø10 cross-bolts, R-clips; 10 safety chains 5/8 in Grade 100 (2 per joint) + 20 Crosby G-2130 1 in shackles (+2 spares).

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Bill of materials - tender T2, coach R2, drawgear

Sheet S-502 · REV L4 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____